

LAKE CHARLES NWS, LA, USA

WMO: 722400

Lat: 30.1253N

Lon: 93.2161W

Elev: 19

StdP: 14.69

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03937

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.4	33.7	16.7	12.8	39.0	21.8	16.4	43.7	23.5	52.3	20.8	52.1	6.6	20	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	95.0	77.6	93.2	77.8	91.7	77.6	81.3	88.5	80.4	87.6	79.9	87.1	7.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.4	152.7	84.2	78.6	148.6	83.6	78.0	145.4	83.2	44.8	88.5	44.0	87.8	43.2	86.9	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.2	18.1	16.3	DB	25.2	98.6	3.4	2.7	22.8	100.6	20.8	102.2	18.8	103.7	16.4	105.7
			WB	22.8	83.2	3.3	1.8	20.4	84.4	18.5	85.5	16.6	86.5	14.2	87.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.3	52.6	56.3	62.3	68.6	76.0	81.5	83.2	83.4	79.5	70.7	61.3	54.7	(6)
	DBStd	12.99	9.70	9.30	8.09	6.39	4.73	2.93	2.49	2.89	4.34	7.05	8.28	9.34	(7)
	HDD50	204	86	40	10	0	0	0	0	0	0	0	11	57	(8)
	HDD65	1387	397	266	146	38	2	0	0	0	0	30	166	342	(9)
	CDD50	7231	166	217	392	559	807	946	1029	1036	885	643	350	201	(10)
	CDD65	2941	13	23	63	147	345	496	564	571	436	208	54	21	(11)
	CDH74	28014	17	40	234	891	2943	5203	6315	6310	4059	1642	299	61	(12)
	CDH80	11045	0	0	13	114	911	2207	2781	2858	1661	474	26	0	(13)
(14) Wind	WSAvg	7.2	8.5	8.7	8.5	8.5	7.3	6.0	5.0	5.0	6.0	6.8	7.4	8.2	(14)
(15) Precipitation	PrecAvg	57.4	5.2	3.4	3.6	3.7	5.5	5.9	5.4	5.7	5.7	4.3	4.2	4.9	(15)
	PrecMax	85.3	14.3	8.0	9.3	13.2	20.7	25.3	13.2	23.6	20.0	21.5	14.1	13.3	(16)
	PrecMin	34.8	0.4	0.4	0.2	0.1	0.0	0.8	0.5	0.0	0.4	0.0	0.1	2.0	(17)
	PrecStd	11.5	3.0	2.0	2.1	3.0	3.9	4.2	3.1	4.2	3.7	4.3	2.6	2.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.1	77.9	81.6	85.4	92.0	96.3	96.2	97.9	96.4	90.4	83.0	78.7	(19)
		MCWB	68.2	68.5	68.4	70.7	74.4	76.9	78.1	78.0	76.1	74.9	71.7	71.1	(20)
	2%	DB	73.4	75.1	79.0	82.9	88.8	92.8	93.9	95.1	92.9	87.5	80.1	75.7	(21)
		MCWB	66.4	67.7	68.9	71.1	74.1	76.7	78.3	78.2	76.2	74.0	70.8	69.9	(22)
	5%	DB	70.7	72.8	76.8	81.1	86.8	91.0	92.3	93.3	90.6	85.1	77.6	72.9	(23)
		MCWB	65.6	66.7	67.8	70.0	74.4	76.9	78.3	78.4	76.0	73.1	69.0	68.1	(24)
	10%	DB	68.1	70.4	74.3	79.2	85.0	89.4	90.8	91.5	88.5	82.5	75.0	70.0	(25)
		MCWB	63.8	65.5	66.8	69.3	74.0	77.1	78.2	78.3	75.6	71.8	67.9	66.1	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.8	72.0	72.9	76.2	80.1	81.5	82.3	82.4	81.6	79.6	75.8	73.7	(27)
		MCDB	73.8	75.2	77.1	80.3	84.7	87.3	90.1	90.0	88.0	84.7	79.1	76.5	(28)
	2%	WB	69.1	70.4	71.3	74.6	78.1	80.3	80.9	81.4	80.0	77.9	73.7	71.3	(29)
		MCDB	71.8	73.3	75.5	79.1	83.9	86.8	88.7	88.9	86.6	83.3	77.0	73.9	(30)
	5%	WB	67.3	68.8	70.3	73.3	76.9	79.5	80.2	80.4	78.8	76.4	72.0	69.4	(31)
		MCDB	69.7	71.5	74.3	77.9	82.9	86.4	87.9	87.9	85.4	81.6	75.6	72.4	(32)
	10%	WB	64.9	66.9	69.0	72.2	75.9	78.7	79.5	79.9	77.9	75.0	69.7	66.8	(33)
		MCDB	67.4	69.4	72.9	76.7	82.0	85.6	87.1	87.3	84.5	79.9	73.9	69.3	(34)
(35) Mean Daily Temperature Range	MDBR		18.5	18.0	18.5	18.3	16.9	15.5	15.1	16.1	17.4	20.0	20.0	18.6	(35)
	5% DB	MCDBR	18.6	17.2	17.3	17.9	17.6	17.5	17.1	18.4	19.3	19.7	20.4	18.7	(36)
		MCWBR	13.3	11.1	9.4	8.5	6.8	5.7	5.6	5.7	6.6	8.7	11.9	13.1	(37)
	5% WB	MCDBR	15.9	13.8	14.5	14.3	14.8	14.4	15.5	16.0	15.6	15.5	17.3	16.6	(38)
		MCWBR	13.2	10.6	9.7	8.6	6.7	5.8	5.8	5.9	6.4	8.8	12.8	13.3	(39)
(40) Clear-Sky Solar Irradiance	taub		0.342	0.358	0.376	0.431	0.466	0.500	0.500	0.488	0.458	0.382	0.367	0.352	(40)
	taud		2.511	2.476	2.419	2.263	2.201	2.141	2.182	2.218	2.268	2.463	2.449	2.484	(41)
	Ebn at Noon		283	287	289	275	265	254	253	256	260	275	271	272	(42)
	Edh at Noon		28	32	36	43	47	49	47	45	41	32	30	28	(43)
(44) All-Sky Solar Radiation	RadAvg		910	1039	1442	1746	1948	1948	1875	1776	1588	1398	1054	808	(44)
	RadStd		90	169	131	131	128	150	146	160	157	174	101	89	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+207	+142
Regional (27 neighbors)		+0.54	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+198	+139

Nomenclature: See separate page